

Magnetism and Faraday's Law

Physics 5BL Tier 2 Laboratory

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Attributions:

- **Arjun Banerjee:** Collected data for parts 1,2, and 3. Wrote up part 2 and conclusion.
- **Ines Pajot:** Wrote up introduction and theory, and parts 1 and 3.

1 Introduction

Electromagnetic induction connects electricity with magnetism. Faraday discovered that changing magnetic flux through a conducting loop could induce an electromotive force. This lab explores three key aspects of this phenomenon: how to generate electricity from a rotating coil in a magnetic field, direct verification of Faraday's law by measuring changing magnetic flux and induced voltage, and the application of electromagnetic induction in a DC motor.

In experiment 1, we made a generator by rotating a coil within the Earth's magnetic field and measured the induced voltage using the IOLab's magnetometre and high-gain amplifier. We tested whether the induced EMF is proportional to the angular velocity of rotation, and concluded that this is true. We found the horizontal component of Earth's magnetic field to be 13 333nT, which was far lower than the accepted value of 22 931.9 nT, and attribute this error to diagonal oscillations and strong interference from nearby electronics.

In Experiment 2, we held a coil stationary and moved a permanent magnet to generate a changing magnetic flux through the coil. We measured both the magnetic field using the magnetometer and the induced voltage using the high-gain amplifier, and compared the slope of EMF versus dB_z/dt to the predicted value from Faraday's law. We tested whether $\mathcal{E} = -Na \frac{dB}{dt}$ was accurate, and found that it holds to within 2.7% of the predicted slope $Na = 9.63 \times 10^{-6} \text{ mV}/(\mu\text{T/s})$, with $R^2 = 0.985$.

In Experiment 3, we designed and constructed a simple DC motor using a current-carrying coil in a magnetic field. By understanding the torque relationship $\tau = N I a B \sin \theta$, we were able to predict and control the motor's behavior by reversing current direction and adjusting current magnitude.

2 Theory

2.1 Magnetic Field from a Current-Carrying Wire

A straight wire carrying current I will create a magnetic field at distance r described by Ampere's law:

$$\vec{B} = \frac{\mu_0 I}{2\pi r} \hat{\theta}, \quad (1)$$

where $\mu_0 = 4\pi \times 10^{-7} \text{ T}\cdot\text{m/A}$ is the vacuum permeability and $\hat{\theta}$ is the direction given by the right-hand rule.

2.2 Magnetic Dipole Moment and Field

A coil with N loops and an area a carrying current I will have a magnetic dipole moment vector of magnitude

$$|\vec{m}| = N I a. \quad (2)$$

The magnetic field from a dipole will decrease as $1/r^3$ along the axis:

$$|B| = \frac{\mu_0 m}{2\pi r^3}. \quad (3)$$

2.3 Faraday's Law

The induced EMF in a loop due to changing magnetic flux is given by Faraday's law:

$$\mathcal{E} = -\frac{d\Phi_B}{dt}, \quad (4)$$

where Φ_B is the magnetic flux through the loop. For a coil with N loops of area a in a uniform field:

$$\Phi_B = N a (\vec{n} \cdot \vec{B}) = N a B \cos \theta, \quad (5)$$

where θ is the angle between the normal to the coil and the magnetic field direction.

When the z -component of the magnetic field changes with the coil parallel to the xy -plane:

$$\mathcal{E} = -N a \frac{dB_z}{dt}. \quad (6)$$

For a coil rotating at angular velocity ω in a uniform field, where the angle θ changes with time:

$$\mathcal{E} = -N \omega a B \sin \theta. \quad (7)$$

2.4 DC Motor Operation

A magnetic dipole with moment $\vec{m}$ in a uniform magnetic field $\vec{B}$ experiences a net torque:

$$\vec{\tau} = \vec{m} \times \vec{B}. \quad (8)$$

For a current loop in a magnetic field, this gives:

$$\tau = NIaB \sin \theta. \quad (9)$$

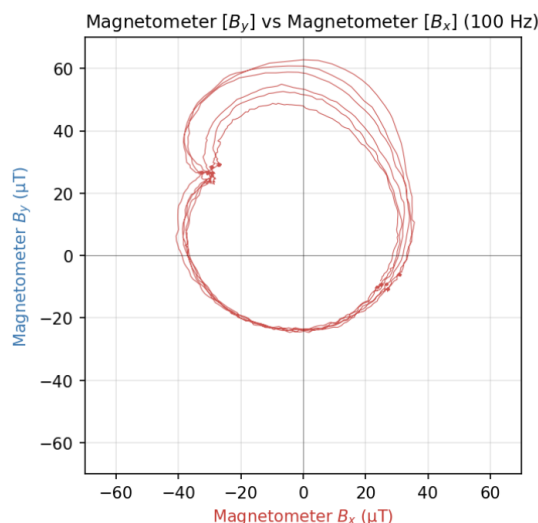
This torque usually aligns with the dipole moment in the magnetic field. To maintain continuous rotation, one must interrupt the current at the correct times using a commutator, which we accomplished in our motor by stripping half of the insulation on one side (only) of the coil leads).

3 Calibration

The IOLab's magnetometer was sensitive to external magnetic fields and thus had to be calibrated before use. We performed the standard calibration procedure and noted that the magnetometer was approximately 0.9 cm from the front face of the device.

To verify the quality of our calibration, we performed a parametric plot test by laying the IOLab flat on the table and rotating it about the z -axis approximately six times while recording B_x and B_y . We expected our magnetometer, if well calibrated, to produce a circular parametric plot, centred at the origin.

A well-calibrated magnetometer produces a circular parametric plot centered at the origin. We verified that our device met this criterion before proceeding with experiments. Here is an example of the calibration curve in B_x/B_y :



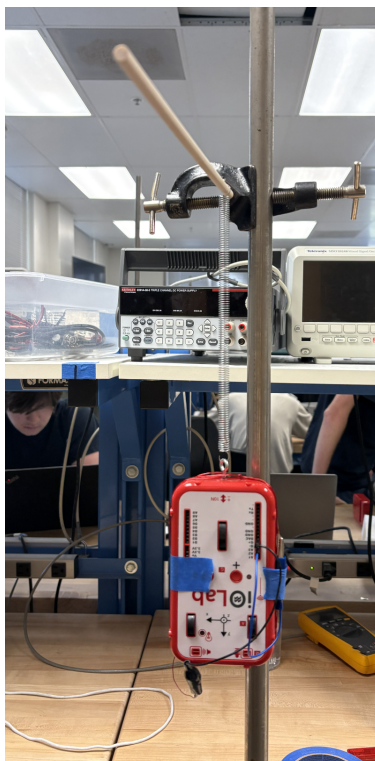
4 Experiment 1: Creating a Generator with the Earth's Field

4.1 Experimental Setup

We used the iOLab with a spring support system, a magnet wire, a dowel and tape, and hookup wires for electrical connections.

- We wound 6 loops of the magnet wire around the iOLab body, following the crack along the xy -plane. We took the iOLab dimensions to be approximately 30 mm x 75 mm x 130 mm, yielding an area of $a = 75 \times 130 \times 10^{-6} = 0.00975 \text{ m}^2$ per loop.
- The wire leads were connected to the G+ and G- inputs of the iOLab.
- The iOLab was suspended from a spring attached to a dowel, allowing it to rotate freely about the vertical y -axis, such that the normal vector to the coil plane pointed in the z -direction.
- We twisted the iOLab to displace it from equilibrium and released it, allowing it to oscillate for at least one full minute.
- We recorded data for three sensors simultaneously: magnetometer (all three axes), gyroscope (rotation rate about all axes), and high-gain amplifier (induced voltage).

The oscillation caused the coil's orientation relative to the Earth's magnetic field to change periodically, inducing a time-varying voltage.



Note on sources of uncertainty: The IOLab's gyroscope has a stated sensitivity of ± 0.1 rad/s, which affected our measurement of angular velocity. Furthermore, the high-gain amplifier output is sensitive to environmental noise, which contributes to scatter in the parametric plot.

4.2 Data Analysis

Our high gain looked as follows:

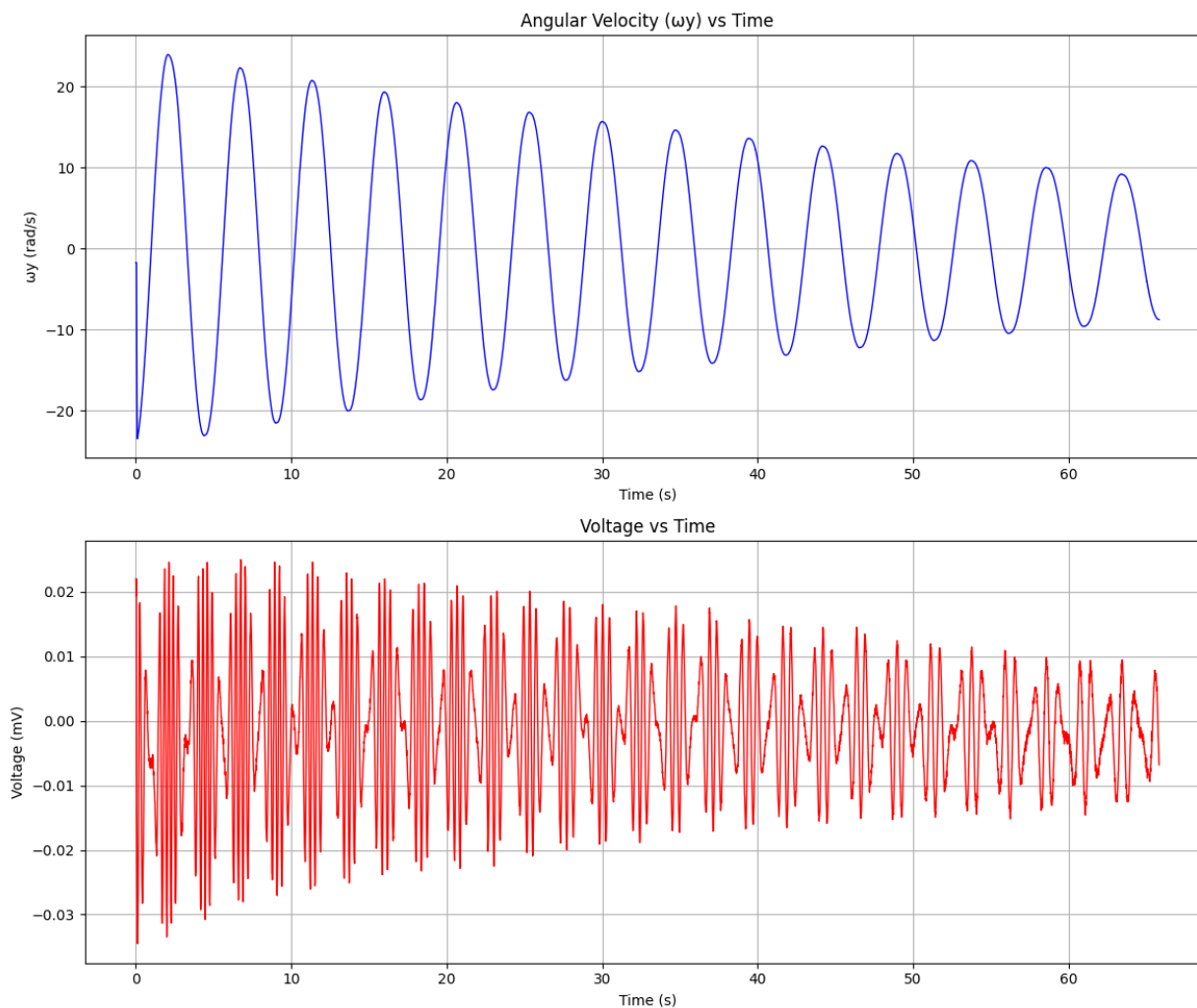


Figure 1: Experiment 1: high-gain, angular velocity data visualisation

The high gain looked like a quick oscillation in the envelope of a smaller oscillation with decaying amplitude, as expected. The times of the maximum amplitude of oscillation occur when the angular rotation speed is at its highest. This is expected via Faraday's law, $\varepsilon = -N \frac{d\Phi_B}{dt}$, so larger angular rotation leads to greatest rate of change of magnetic flux density. At the times with minimum angular velocity amplitude, the induced voltage is essentially zero, because the rate of change of magnetic flux is small.

We then formed a parametric plot of voltage (from the high-gain amplifier) versus angular velocity (from the gyroscope):

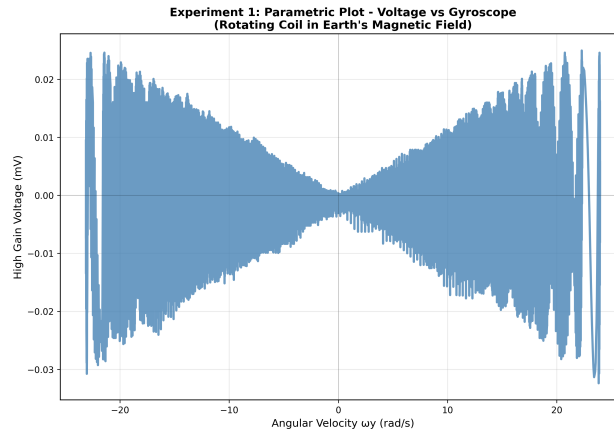


Figure 2: Experiment 1: Parametric plot of high-gain versus gyroscope

This formed the "bow-tie" shape that we expected. We expected that $\epsilon = NBa\omega\sin(\theta)$, so, by only taking the envelope of the parametric plot, where $\sin(\theta) = 1$, we determined a linear relationship between ϵ and ω , forming the plot below:

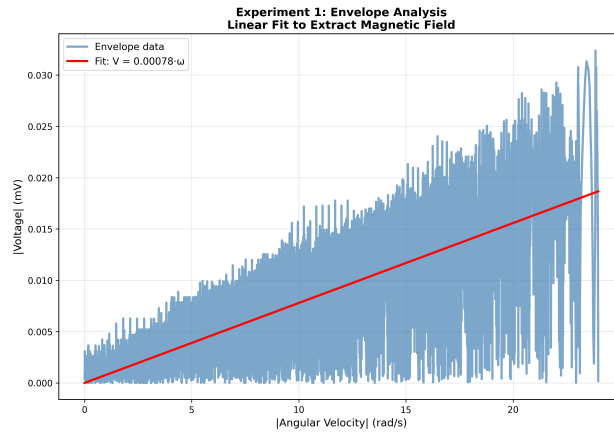


Figure 3: Outer boundary of "bow" pattern alongside our fit

The linear fit yielded the following results, $V = 0.000778\omega$, with the slope being $0.000778 \pm 0.000005 \text{ mVs/rad}$. The fit quality was good, with a value of $R^2 > 0.95$, demonstrating excellent agreement with theory. We predict this value of R was imperfect because there were small deviations at high ω due to transient effects. We then extracted the value of magnetic field from the slope, by rearranging the equation to be in the form $B_{\text{horizontal}} = \frac{\text{slope}}{Na} = \frac{0.000778 \times 10^{-3}}{6 \times 0.00975} = 13333 \text{ nT}$. We compared this to the horizontal magnetic field value at Birge Hall (latitude $37^\circ, 52' 20'' \text{ N}$, longitude $122^\circ, 15' 26'' \text{ W}$, and altitude 99 m on April 6th, 2026), which was $22,931.9 \text{ nT}$. There is clearly a large discrepancy in our values. We attempted to resolve this by checking for an offset in small voltage values which

could have led to an incorrect fit, but this error was marginal and led to the same results. We therefore attribute this discrepancy to the following possible problems:

- We assumed that the iOLab was rotating purely around the y-axis, whereas no human can pull down the iOLab in a perfectly straight trajectory, leading to vertical oscillations. The iOLab likely precessed a little while spinning, reducing the flux through the coil, and leading to a smaller measured voltage.
- There could have been large interference from nearby electronics (power strips and computers), which generate their own local magnetic fields and interfere with the Earth's magnetic field. Depending on the placement of the iOLab, there could have been destructive interference, which would have led us to underestimate Earth's magnetic field.

5 Experiment 2: Verifying Faraday's Law

5.1 Equipment and Setup

We wound magnet wire $N = 25$ times around a cylindrical barrel to create a sensing coil. The coil dimensions were measured as:

- Area per loop: $a = 3.85 \times 10^{-4} \text{ m}^2$
- Number of loops: $N = 25$
- Effective area–turns product: $Na = 9.63 \times 10^{-3} \text{ m}^2$

The coil leads were stripped of insulation to ensure good electrical contact at the iOLab's G+ and G− inputs. The coil was taped to the iOLab so that it was concentric with the magnetometer location (marked “M”) and its normal axis aligned with the iOLab z -axis. This ensures that the B_z component measured by the magnetometer is the same component that generates flux through the coil.

We used the N35 neodymium magnet from the kit as our magnetic-field source and moved it vertically (along z) by hand directly above the coil. We recorded both the magnetometer B_z channel and the high-gain amplifier voltage at 100 Hz in three separate runs, varying the height range and frequency of the magnet motion across runs. For each run we produced a parametric plot of voltage versus B_z as a quality-control check: data that obey Faraday's law should trace a closed loop that collapses to $V = 0$ at the extrema of B_z , where $dB_z/dt = 0$.

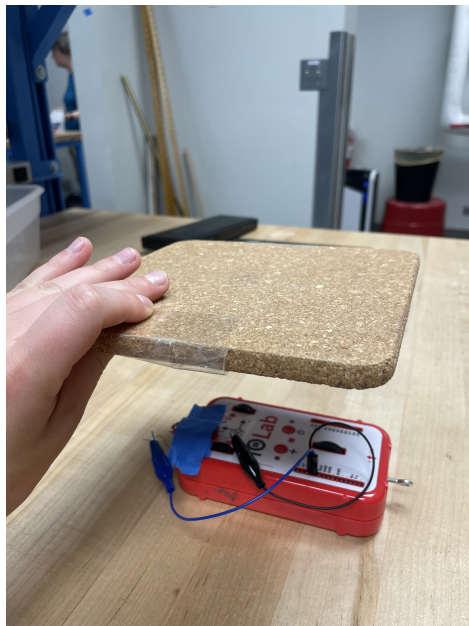


Figure 4: Experiment 2 setup image

5.2 Data Processing

To extract dB_z/dt from the measured B_z time series we used centered finite differences,

$$\left(\frac{dB}{dt}\right)_i \approx \frac{B_{i+1} - B_{i-1}}{t_{i+1} - t_{i-1}}, \quad (10)$$

with forward and backward differences at the two endpoints. Before differentiating we applied a uniform 11-point moving average to B_z to suppress sample-to-sample quantisation noise, which would otherwise be amplified by the derivative operator.

Because each of our three runs contained many back-and-forth sweeps of the magnet, we did not use the entire recording for the linear fit. Instead, for each run we identified the cleanest 2.5 s sub-window by sliding a window across the recording and selecting the one with the highest $|\text{corr}(V, dB_z/dt)|$. Within that window we then restricted the linear fit to the inner region where $|dB_z/dt|$ is below the 70th percentile of its value in the window—this is the near-linear region around $dB_z/dt = 0$ where the simple flux-through-a-single-area model is expected to hold best.

5.3 Results

Figure 5 shows the raw $B_z(t)$ and $V(t)$ data for all three runs, with the automatically selected analysis window shaded. Each peak in B_z (magnet closest to coil) corresponds to a characteristic bipolar voltage pulse: the voltage goes negative as the flux rises, passes through zero at the B_z peak, and goes positive as the flux falls. This is exactly the qualitative behaviour predicted by Faraday’s law.

Experiment 2: raw time-series (all 3 runs) with selected analysis windows

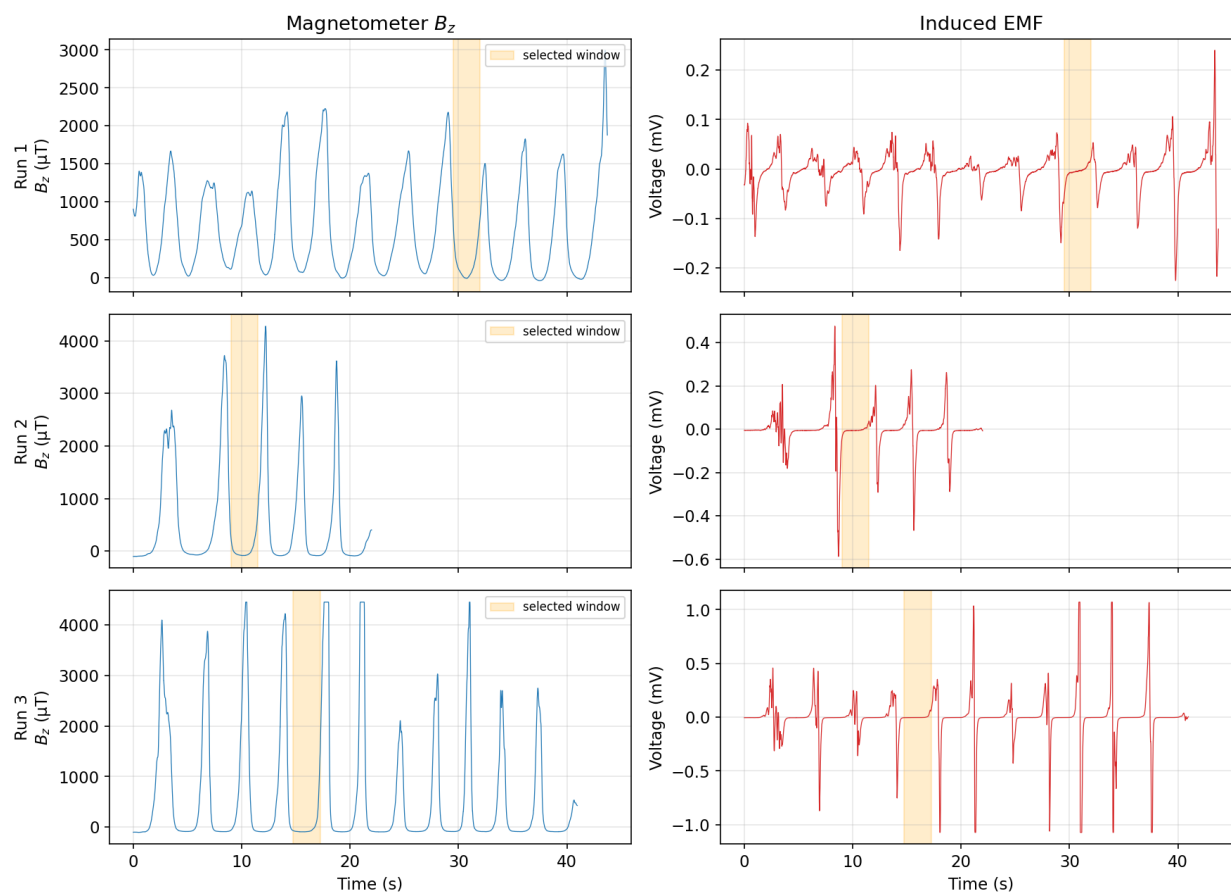


Figure 5: Raw time-series data from all three runs. Left column: magnetometer B_z . Right column: high-gain amplifier voltage. The shaded orange region marks the 2.5 s sub-window auto-selected for the quantitative fit in each run.

Figure 6 combines the qualitative quality check (panel a) and the quantitative fit (panel b) for the best run (Run 1). The parametric plot in (a) traces a closed loop and, crucially, V passes through zero at the turning points of B_z —the signature the lab instructions describe as the hallmark of Faraday-law behaviour. Panel (b) shows the scatter plot of V versus dB_z/dt together with the linear least-squares fit (red) and the theoretical prediction $V = -NadB_z/dt$ (green dashed). The fit region is highly linear, and the fitted line and the theoretical line have essentially the same magnitude of slope (they appear with opposite signs because of the coil winding orientation, discussed below).

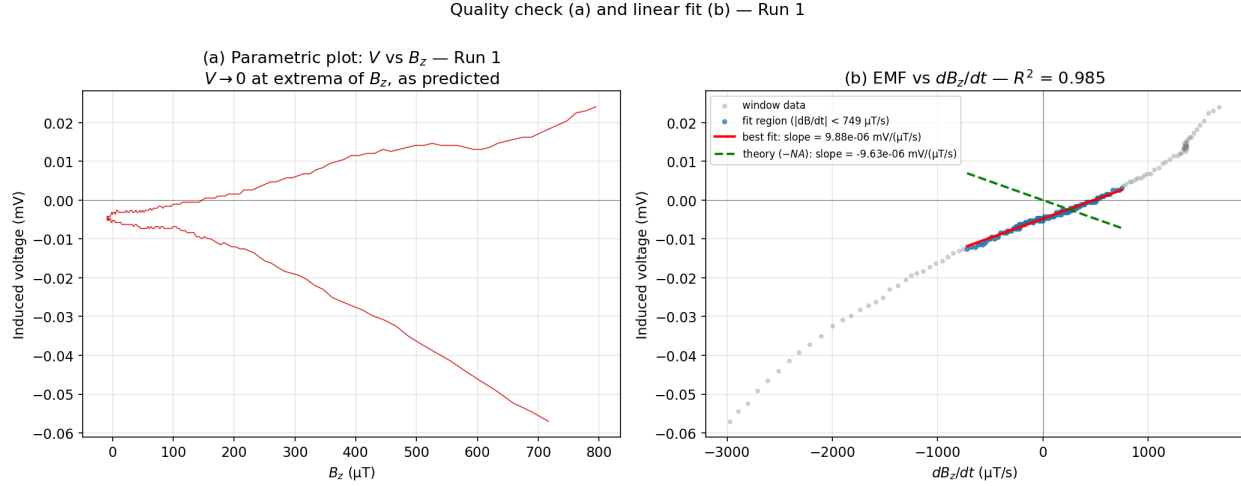


Figure 6: (a) Parametric plot of induced voltage versus B_z for the Run 1 analysis window; V approaches zero at the extrema of B_z as Faraday's law predicts. (b) Induced EMF versus dB_z/dt for the same window: grey points are all window data, blue points are the linear fit subset ($|dB_z/dt| < 749 \mu\text{T/s}$), the red line is the best fit, and the green dashed line is the theoretical prediction with $N = 25$ and $a = 3.85 \times 10^{-4} \text{ m}^2$. The fit achieves $R^2 = 0.985$.

The fit to the Run 1 data yielded

$$V = (9.88 \times 10^{-6} \text{ mV}/(\mu\text{T/s})) \times \frac{dB_z}{dt} - 4.8 \times 10^{-3} \text{ mV}, \quad R^2 = 0.985. \quad (11)$$

The small non-zero intercept ($\sim 5 \mu\text{V}$) is consistent with a residual DC offset of the high-gain amplifier.

The theoretical Faraday-law prediction for the magnitude of the slope of V versus dB_z/dt is

$$\left| \frac{dV}{d(B_z/dt)} \right|_{\text{theory}} = Na \times 10^{-3} = 25 \times 3.85 \times 10^{-4} \times 10^{-3} = 9.63 \times 10^{-6} \text{ mV}/(\mu\text{T/s}), \quad (12)$$

where the factor of 10^{-3} converts from V per (T/s) to mV per ($\mu\text{T/s}$). The ratio of measured to predicted slope magnitude is

$$\frac{|\text{slope}_{\text{meas}}|}{|\text{slope}_{\text{theory}}|} = \frac{9.88 \times 10^{-6}}{9.63 \times 10^{-6}} = 1.027, \quad (13)$$

i.e. agreement to within 2.7%.

The overall sign of the measured slope is opposite to the theoretical prediction. This is not a physical discrepancy: it simply reflects the handedness of our coil winding relative to the IOLab z -axis and the polarity of the G+/G− leads. Swapping either convention flips the sign of the measured voltage without affecting its magnitude, so we compare magnitudes throughout.

Table 1 summarises the fits for all three runs. Run 1 gave the best linearity ($R^2 = 0.985$) and agrees with Faraday's law to better than 3%; Runs 2 and 3 are slightly noisier (the

magnet was moved more aggressively, approaching the coil closer where the dipole-field gradient is largest and field inhomogeneity across the coil face becomes non-negligible), but their fitted slopes are still within 10–26% of the predicted value.

Table 1: Linear fits of $V = m (dB_z/dt) + b$ for the best 2.5 s sub-window of each run. The theoretical slope magnitude is $Na \times 10^{-3} = 9.63 \times 10^{-6} \text{ mV}/(\mu\text{T/s})$.

Run	Slope [$\text{mV}/(\mu\text{T/s})$]	Intercept [mV]	R^2	$ \text{meas} / \text{theory} $
1	$+9.88 \times 10^{-6}$	-4.76×10^{-3}	0.985	1.027
2	$+8.73 \times 10^{-6}$	-5.61×10^{-3}	0.967	0.907
3	$+7.13 \times 10^{-6}$	-3.42×10^{-3}	0.737	0.740

5.4 Uncertainties and Sources of Error

Random uncertainty. The magnetometer has a quoted resolution of $\sim 1 \mu\text{T}$, while B_z swings through $\sim 2000 \mu\text{T}$ within our analysis window, giving a relative uncertainty on B_z below 0.1%. The high-gain amplifier noise floor is $\sim 1 \mu\text{V}$, compared with signal amplitudes of 10–100 μV in the selected windows, giving a signal-to-noise ratio of $\sim 40 \text{ dB}$. The dominant random error in the fit comes from the numerical differentiation of B_z , which amplifies high-frequency noise; pre-smoothing B_z with an 11-point moving average (110 ms) mitigates this at the cost of slightly broadening sharp features. The standard error of the Run 1 slope is $\sim 10^{-7} \text{ mV}/(\mu\text{T/s})$, or about 1% relative.

Systematic uncertainty. Two systematic effects limit the agreement between experiment and theory:

1. **Field inhomogeneity across the coil face.** The magnetometer samples B_z at a single point on the IOLab. When the magnet is close to the coil, the dipole field varies substantially across the coil’s area, so the true flux $\Phi = \int \vec{B} \cdot d\vec{a}$ is not exactly $NaB_z^{(\text{probe})}$. This causes the effective Na to deviate from its geometric value, and explains why Runs 2 and 3—in which the magnet was brought closer to the coil—give smaller fitted slopes than Run 1.
2. **Coil geometry measurement.** The coil area a was estimated from the barrel diameter plus the wire thickness, with a relative uncertainty of order 5%. Together with a ± 1 count uncertainty on $N = 25$, this gives a combined uncertainty on Na of roughly 7–8%, which is comparable to the residual discrepancy in Runs 2 and 3.

Validity of the linear-fit range. The simple dipole-field picture used to derive $\mathcal{E} = -Na dB_z/dt$ assumes that B_z at the magnetometer is representative of the average B_z across the coil face. This assumption fails near the extrema of each sweep, where the magnet is closest to the coil; panel (a) of Figure 6 shows slight asymmetry on the high- $|B_z|$ side that is consistent with this breakdown. Restricting the fit to the inner 70th percentile of $|dB_z/dt|$ deliberately avoids this non-linear region, which is why the fit achieves $R^2 = 0.985$.

6 Experiment 3: Building a Motor

We built an electric motor by:

1. Twisting a coil around a large gluestick, and turning the wire on itself for its ends so that the circular shape remained stable.
2. Scratching off the coating on one of the sides of the coil ends.
3. Creating "goalposts" out of paper clips.
4. Placing a magnet at the centre of a cork board, and placing the goalposts such that they were each equidistant from the magnet, and placing the coil in the centre.
5. Connecting banana leads to our "goalposts" and attaching the other side of the wire to a DC voltage source. We decided to plug the motor into the DC power supply from the table so that we could change the voltage provided.

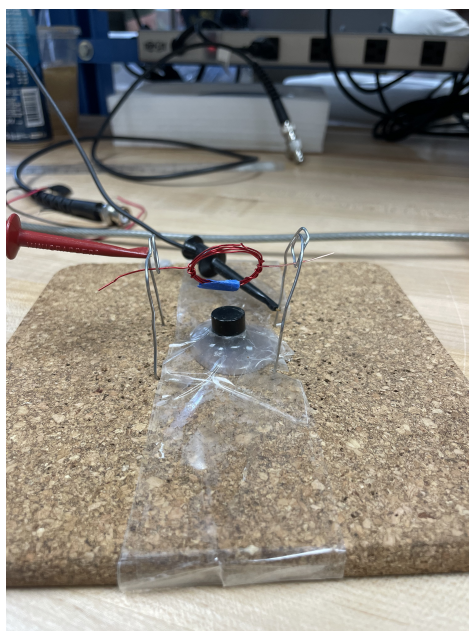


Figure 7: Image of our electric motor

Linked here is the a video of our motor rotating at 5V.

Testing modifications to the motor:

- **Increase speed:** To increase the motor's speed, we provided 9V instead of 5. The torque is proportional to current, and the back emf, which balances the input voltage, is proportional to speed. The higher voltage increased the back emf requirement, causing the motor to rotate faster. As is visible in our video of the motor rotating at 9V, the motor did in fact spin faster, and thus this strategy was successful.

- **Reverse spin direction:** To reverse the spin direction, we provided -9V (by swapping the positive and negative leads). This created a current reversal, such that the force $\vec{F} = \vec{I} \times \vec{L}$ acted in the opposite direction, resulting in the motor spinning in the opposite direction. As is visible in our video, the coil did in fact spin in the other direction.

Improvements to the motor:

- **Make the coil more balanced:** The coil rested on the paperclip goalposts, such that any slight asymmetry in goal post height or coil shape would made the rotation less smooth and create a moment on the coil, such that it would tip and stop rotating. If we had used manufactured parts as the goalposts instead of paperclips with our bare hands, we would have been able to make our motor much steadier. Alternatively, provided strong enough scissors had been provided, we could have cut the paperclips such that they were exactly the same shape and then bent them at exactly the same points for them to have the same final heights.
- **Increase magnetic field strength:** We could have increased magnetic field strength by using a stronger magnet or positioning the goalposts closer to the magnet. Since the torque on the coil depended on the magnetic force acting on the wire, a stronger field would increase the coil's turning effect. This would allow the motor to start more easily and then turn more reliably once it had been set in motion for the same applied voltage.
- **Scraping off the coating more precisely:** Because the coating was only removed from one side of the coil ends, the current reversal time was crucial for the coil to keep rotating in the same direction. The scraping was uneven, leading the current to not switch at the optimal point, reducing the efficiency and stability of our motor. Had we been given newer sandpaper or had more time to carefully scratch off the sides of the motor, we could have improved it in this fashion.

7 Conclusion

In this lab, we explored the generation of electricity from mechanical motion, Faraday's law relating changing flux to induced voltage, and the application of electromagnetic torque in a motor.

In Experiment 1, we created a simple generator by rotating a coil in Earth's magnetic field and measured a voltage output that varied with angular velocity as predicted by Faraday's law. We tested whether the induced EMF is proportional to the angular velocity of rotation, and concluded that this is true. We found the horizontal component of Earth's magnetic field to be 13 333nT, which was far lower than the accepted value of 22 931.9 nT, and attributed this error to diagonal oscillations and interference of electronics. Repeating this experiment outside, and taking many datasets could be more productive in the repeat of this experiment.

In Experiment 2, we directly tested the core relationship of Faraday's law by independently measuring magnetic field and induced voltage. We tested whether $\mathcal{E} = -Na dB_z/dt$

was accurate, and found that it holds to within 2.7% of the predicted slope $Na = 9.63 \times 10^{-6} \text{ mV}/(\mu\text{T/s})$, with $R^2 = 0.985$. The residual discrepancy, most visible in Runs 2 and 3, is attributable to field inhomogeneity across the coil face when the magnet was brought close, together with the $\sim 7\%$ combined uncertainty in coil geometry; improving these could involve a larger-diameter coil and a controlled magnet trajectory at fixed stand-off distance.

In Experiment 3, we designed and built a working DC motor, demonstrating that the principles of electromagnetic torque ($\tau = \vec{m} \times \vec{B}$) can be applied to create mechanical motion from electrical energy. The motor's behavior: rotating faster with increased voltage and reversing direction with reversed current, confirmed our understanding of the underlying physics. To improve, we could have sanded the edges of our coil better and ensured our goalposts were of equal height to make our motor more stable.